

# LUC MCCUTCHEON

SENIOR RESEARCH SCIENTIST

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[lucmccutcheon.home@gmail.com](mailto:lucmccutcheon.home@gmail.com) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) Senior Research Scientist specialising in Reinforcement Learning, world models and distributed post-training for embodied robotics. Experience deploying full-stack AI solutions and leading technical teams. Expert in coding algorithms from scratch in PyTorch/JAX, with published research (ICRA, NeurIPS, IROS) focusing on real-world embodied AI challenges.

## RESEARCH SCIENTIST (FIXED-TERM)

Cambridge Consultants · Oct 2025 – Apr 2026

- Led the Reinforcement Learning workstream for the Unitree G1 humanoid, delivering robust policies for dynamic environments.
- Engineered an RL fine-tuning pipeline for VLA models (GR00T 1.6) using PPO/GRPO within custom Isaac Sim environments.
- Designed MuJoCo and Isaac Sim environments supporting curriculum and continual learning to minimise the sim-to-real gap.
- Coupled individual foot control with MPC planners to achieve precise, stability-aware foot placement in 3D space.
- Executed sim-to-real transfer via ONNX to deploy blind stair traversal locomotion on physical hardware.
- Trained cooperative multi-agent RL policies for a safety-critical defence application, using Kubernetes for distributed training.

## LEAD RESEARCH SCIENTIST (PART-TIME)

Agile Loop · Oct 2023 – Apr 2025

- Led a team of 10 researchers developing autonomous AI agents, driving technical recruitment and presenting to VC firms.
- Architected agentic pipelines from scratch and steered the internal research strategy, ranging from fundamental deep learning architecture to RL and VLM LoRA fine-tuning.
- Presented our full-stack approach to the Google Cloud team during their workshop for AI agents.

## RESEARCH SCIENTIST (PART-TIME)

Agile Loop · Jun 2023 – Oct 2023

- Pioneered RL environments for software and web-apps, using computer vision for icon recognition, and creating systems for distributed multi-task training.
- Developed software and infrastructure allowing hybrid edge/cloud model routing based on task complexity for Lenovo POC.

## RESEARCH ASSISTANT (VOLUNTEER)

Connected & Autonomous Vehicle Lab · Jun 2022 – Aug 2022

- Improved exploration using Noisy Neural Networks and tackled partial observability using LSTMs and Deep Q-Learning.

## SOFTWARE ENGINEER INTERN

QinetiQ · Jul 2018 – Sep 2018 & Jul 2019 – Sep 2019

- Engineered tactical voice cryptography (C++/Python) and automated red team attacks across two placements, bridging offensive and defensive security operations.

## PHD REINFORCEMENT LEARNING (SPONSORED BY VEOLIA NUCLEAR SOLUTIONS)

University of Surrey · 2021 – present

- Designed and implemented time-delay mitigating controller for Veolia Nuclear Solutions' local-remote manipulator (DEXTER).
- Learned a probabilistic World Model to “undelay” RL environments to provide adaptive PD gains in highly stochastic conditions.

- Improved stability analysis through learning a Lyapunov function, using RL to generate counter-examples improving robustness.
- Co-designed and lectured “Intelligent Vehicle Design” university course, teaching planning methods for robotic control.
- Implemented various papers and algorithms from scratch in JAX, such as PCGrad, PPO, ReDo, CBP, VAE, ResNet and many more.
- Implemented numerous custom RL environments, creating complex wrappers for random time delayed environments and local-remote system parallelism.

## BSC COMPUTER SCIENCE (HONS), FIRST CLASS

University of Surrey · 2018 – 2021

- Recipient of the prestigious CyberFirst Scholarship, sponsored by QinetiQ.
- 1st Class Degree, key modules: Artificial Intelligence (92%), Data Structures & Algorithms (86%), Computational Intelligence (83%).
- Dissertation in time-series financial forecasting using an LSTM Neural Network with various financial indicators.
- Key takeaways: CUDA programming, evolutionary algorithms and fundamental mathematics.

**Frameworks:** JAX, PyTorch, Gymnasium, vLLM, Raylib

**Code:** Python, C++, Rust, JavaScript, Lua, Bash

**Methods:** World models, PPO/GRPO, SAC, LoRA, VLA, Sim2Real

**Compute / Infrastructure:** Unitree G1, SLURM, GCP, AWS, Docker

**Languages:** English (Native), French (C1)

### First Author

- **Calibrated Partial Resets: Preventing Policy Collapse in Continual Reinforcement Learning** · Under Review. Implemented PPO, Continual Backpropagation (CBP), Recycling Dormant Neurons (ReDo) and a custom optimiser (CPR). Identifies parameter resets as a source of policy collapse in continual RL and improves stability through soft parameter transformations.
- **Neural Lyapunov Function Approximation with Self-Supervised Reinforcement Learning** · ICRA 2025. Efficient probabilistic world model for stable RL under stochastic conditions, jointly learning a SAC control policy, world model and Lyapunov function and exploiting geometric properties for stability analysis.
- **Adaptive PD Control using Deep RL for Local-Remote Teleoperation with Stochastic Time Delays** · IROS 2023. Proposed PMDC, a model-based RL framework using a learned world model ensemble to mitigate stochastic delays up to ~300ms.

### Co-Author

- **Meta-World+: An Improved, Standardized, RL Benchmark** · NeurIPS 2025 & ICML 2025 Workshop (Spotlight). Implemented optimiser algorithms such as PCGrad and GradNorm for multi-task learning.
- **Exploring In-Context Ensemble with Video-Language Models for Low-Level Workflow Understanding** · NeurIPS 2024 Workshop. Project definition and implementation of Gemini models into a novel in-context ensemble learning framework.
- **Prediction Based Decision Making for Autonomous Highway Driving** · ITSC 2022. Implemented the “Noisy Networks” baseline and part of the Rainbow algorithm.

All implementations are created from scratch using PyTorch or JAX/Flax and trained on a SLURM cluster or Google Cloud, each demonstrating state-of-the-art performance against baselines on challenging benchmarks.

### Reviewing

ICLR 2026 | NeurIPS 2025 | TNNLS 2025 | IROS 2025

- 3rd Place: Complex Worlds (General Reasoning) Hackathon
- Honourable Mention: (xAI) Grokathon 2026
- Bronze: Mathematics Olympiad
- Bronze: British Informatics Olympiad
- Grace Hopper Award: Computer Science
- Foundership Award: Student Enterprise
- NVIDIA Fundamentals of Deep Learning
- Coursera Reinforcement Learning Specialisation
- **Guest Speaker & Panel Discussion (Agents Workshop)** · Google
- **Model-based Reinforcement Learning** · UoS School of Engineering
- **JAX vs PyTorch** · UoS School of Engineering
- **Model Optimisation & Compilers** · Cambridge Consultants
- **Safe Reinforcement Learning** · Cambridge Consultants